THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An atom has the configuration 2, 8, 6. The number of protons is: 1

(a) 16 **(b)** 8 **(c)** 14 **(d)** 6

2. An element has 3 shells and 2 valence electrons. Its atomic number is: 1

(a) 12 **(b)** 10 **(c)** 14 **(d)** 8

3. Chlorine has isotopes of mass 35 u (75 per cent) and 37 u (25 per cent). The average atomic mass is: 1

(a) 36 u **(b)** 35 u **(c)** 37 u **(d)** 35.5 u

4. An element has atomic number 19. The number of valence electrons is: 1

(a) 2 **(b)** 9 **(c)** 8 **(d)** 1

5. An element with atomic number 14 has valency: 1

(a) 8 **(b)** 3 **(c)** 2 **(d)** 4

6. Boron has isotopes 10 (20 per cent) and 11 (80 per cent). The average atomic mass is: 1

(a) 10.5 u **(b)** 11 u **(c)** 10.2 u **(d)** 10.8 u

7. The atom with the configuration 2, 8, 8, 2 is: 1

(a) potassium **(b)** argon **(c)** calcium **(d)** magnesium

8. Which element has a valency of 3? 1

(a) sodium **(b)** aluminium **(c)** chlorine **(d)** magnesium

9. The isotopes of an element differ in the number of: 1

(a) protons **(b)** shells **(c)** electrons **(d)** neutrons

10. The ion Mg^{2+} has the electronic configuration: 1

(a) 2, 8, 8 **(b)** 2, 8 **(c)** 2, 6 **(d)** 2, 8, 2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. An atom has 12 protons and 12 neutrons. Find its mass number and electronic configuration. 2

12. Calculate the average atomic mass of an element with isotopes 63 u (70 per cent) and 65 u (30 per cent). 2

13. The element X has valency 2 and its atom has 12 neutrons and mass number 24. Find the atomic number and the electronic configuration. 2

14. Why do isotopes of an element show similar chemical properties but different physical properties? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. An element has 20 protons and 20 neutrons. Find its mass number, electronic configuration, valency and name. 3

16. An atom has 3 shells. The valence shell has 3 electrons. Find the atomic number and the valency of the element. 3

17. Explain how the Bohr model accounts for the atomic spectra, in brief. 3

OR

Complete: $^{35}_{17}\mathrm{Cl}$, $^{37}_{17}\mathrm{Cl}$: number of p, n, e in each. What are they called? 3

18. Explain how radioactive iodine and cobalt help in medicine. 3

OR

Explain why helium is placed with the noble gases though it has 2 valence electrons. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. An atom X has atomic number 13 and mass number 27. Find the electrons, protons, neutrons, configuration, valency, the ion it forms and the formula of its oxide. 5

OR

Describe the isotopes of carbon, their composition, use and the idea of carbon dating. 5

20. Chlorine has isotopes of 35 and 37 u. If its average atomic mass is 35.5 u find the percentage of each isotope, and explain the reason. 5

OR

Explain, giving reasons, why isotopes cannot be separated by chemical methods but by physical methods such as mass differences. 5

END OF QUESTION PAPER — Total: 40 Marks

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